

Rational equations

Clear the denominators, solve, then throw away the roots the original equation never allowed.

LESSON 28–29 2 × 40 MIN ALGEBRA 10, §2.1 P1 · 1.7 (EXTENSION)

LESSON CLOCK	LEARNING OBJECTIVES	TEXTBOOK
8' 16' 16' 18' 22' 10'	- State the domain of a rational equation before solving it. - Solve an equation by finding the lowest common denominator. - Identify and reject extraneous roots. - Solve equations that become quadratic after a substitution.	National pp. 94–103 Cambridge Pure Mathematics 1, pp. 18–23 Workbook P1 Exercise 1F
Why a check is compulsory here 8 min The method 16 min Extraneous roots 16 min Disguised quadratics 18 min Practice 22 min Homework 10 min		

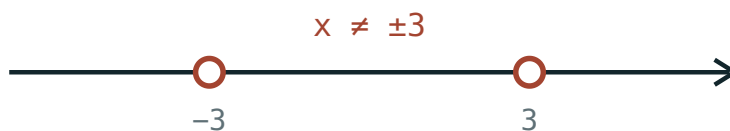
01 Explanation

The domain comes first

WRITE IT BEFORE YOU SOLVE

Every denominator gives a forbidden value. Write $x \neq \dots$ on the first line, before any algebra. It is not a formality: it is the only thing that will tell you which of your final answers is real.

For $\frac{3}{x-2} + \frac{1}{x} = 2$ the domain is $x \neq 0$ and $x \neq 2$.



The two excluded points, marked as open circles before any solving begins.

The method

1. State the domain.
2. Factorise every denominator, and find the lowest common denominator.
3. Multiply every term by the LCD, and cancel.
4. Solve the polynomial equation that results.
5. Reject any root that is outside the domain.

$$\frac{3}{x-2} + \frac{1}{x} = 2 \Rightarrow 3x + (x-2) = 2x(x-2)$$

$$4x - 2 = 2x^2 - 4x \Rightarrow 2x^2 - 8x + 2 = 0 \Rightarrow x^2 - 4x + 1 = 0 \Rightarrow x = 2 \pm \sqrt{3}$$

Both roots are allowed, so both are answers.

MULTIPLY EVERY TERM

A term with no fraction is multiplied by the LCD too. Forgetting the right-hand side is the commonest arithmetic error in the topic.

Extraneous roots — where they come from

Multiplying by $x - 3$ is only reversible when $x \neq 3$. If it happens to be zero, you have multiplied both sides by zero, and the new equation is **not equivalent** to the old one — it has extra solutions the original never had.

$$\frac{x^2}{x-3} = \frac{9}{x-3} \Rightarrow x^2 = 9 \Rightarrow x = \pm 3$$

But $x = 3$ is outside the domain. The only solution is $x = -3$.

THE CHECK IS PART OF THE ANSWER

Never write a final answer without testing it against the domain line you wrote at the start. Examiners award a mark for the rejection, and take one for its absence.

Disguised quadratics

When the same expression appears twice, name it and the equation becomes quadratic:

$$x^2 + \frac{1}{x^2} = 2.5, \text{ let } t = x + \frac{1}{x}$$

Then $t^2 = x^2 + 2 + \frac{1}{x^2}$, so the equation reads $t^2 - 2 = 2.5$, giving $t = \pm 1.5$ — two much easier equations to finish.

The other standard shape is $\frac{x}{x+1} + \frac{x+1}{x} = 2.5$: set $t = \frac{x}{x+1}$ and it becomes $t + \frac{1}{t} = 2.5$.

02 Worked examples

EXAMPLE 1 Solve $\frac{2}{x} + \frac{3}{x+1} = 2$.

① Domain $x \neq 0, -1$.

② LCD $x(x+1)$.

$$2(x+1) + 3x = 2x(x+1)$$

③ $5x + 2 = 2x^2 + 2x$

④ $2x^2 - 3x - 2 = 0 \Rightarrow (2x+1)(x-2) = 0$

Both allowed.

ANSWER

$$x = 2 \text{ or } x = -0.5$$

EXAMPLE 2 Solve $\frac{x}{x-1} - \frac{1}{x-1} = 3$.

① Domain $x \neq 1$.

② Same denominator: $\frac{x-1}{x-1} = 1$.

For every allowed x .

③ $1 = 3$

A contradiction.

ANSWER

No solution

EXAMPLE 3 Solve $\frac{x^2}{x-2} = \frac{4}{x-2}$.

① Domain $x \neq 2$.

② $x^2 = 4 \Rightarrow x = \pm 2$

③ $x = 2$ is excluded.

ANSWER

$x = -2$ only

03 Interactive model

Show the graph of each side and let the class see where the removed point sits.

Domain first

Domain of $\frac{1}{x-5} = 2$:

Solving $\frac{x^2}{x-3} = \frac{9}{x-3}$ gives:

An extraneous root appears because:

Solve $\frac{2}{x} = 1$:

$x = 2$

$x = 0.5$

$x = 0$

no solution

For $\frac{1}{x} + \frac{1}{x-1} = 5$ the LCD is:

x

$x - 1$

$x(x - 1)$

x^2

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Rational equation	Ratsional tenglama	Рациональное уравнение
Denominator	Maxraj	Знаменатель
Lowest common denominator	Eng kichik umumiy maxraj	Наименьший общий знаменатель
Extraneous root	Chet ildiz	Посторонний корень
Domain restriction	Aniqlanish sohasi cheklovi	Ограничение области определения
Clearing fractions	Kasrlardan qutulish	Освобождение от дробей
Equivalent equations	Teng kuchli tenglamalar	Равносильные уравнения
Check (substitution)	Tekshirish	Проверка
Substitution variable	Yordamchi o'zgaruvchi	Вспомогательная переменная

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The first line of a rational equation should be:

the LCD

the domain

the answer

a graph

Multiplying by the LCD may introduce:

lost roots

extraneous roots

nothing

fractions

$x - 1$ $x - 1$ equals 1:

always

for $x \neq 1$

never

only at $x = 1$

For $x^2 + \underline{1} x^2 = 2.5$ a good substitution is:

$$t = x^2$$

$$t = x + \frac{1}{x}$$

$$t = x - 1$$

none

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Domain of $\frac{1}{x-4} = 3$

ANSWER $x \neq 4$

2. Solve $\frac{6}{x} = 2$

ANSWER $x = 3$

3. Solve $\frac{1}{x+1} = 1$

ANSWER $x = 0$

4. Solve $\frac{x}{2} = \frac{3}{x}$

ANSWER $x = \pm\sqrt{6}$

5. Domain of $\frac{1}{x} + \frac{1}{x-2}$

ANSWER $x \neq 0, 2$

6. LCD of $\frac{1}{x}$ and $\frac{1}{x+3}$

ANSWER $x(x+3)$

7. Solve $\frac{4}{x-1} = 2$

ANSWER $x = 3$

Medium · the standard question

7 PROBLEMS

1. Solve $\frac{2}{x} + \frac{3}{x+1} = 2$

ANSWER $x = 2, -0.5$

2. Solve $\frac{x}{x+2} = \frac{3}{x}$

ANSWER $x = 6$ or $x = -1$

3. Solve $\frac{1}{x-1} + \frac{1}{x+1} = 1$

ANSWER $x = 1 \pm \sqrt{3}$

4. Solve $\frac{x^2}{x-2} = \frac{4}{x-2}$

ANSWER $x = -2$

5. Solve $\frac{3}{x} - \frac{2}{x^2} = 1$

ANSWER $x = 1, 2$

6. Solve $\frac{x+1}{x-1} = \frac{x-1}{x+1}$

ANSWER $x = 0$

7. Solve $\frac{x}{x-3} - \frac{3}{x-3} = 2$

ANSWER no solution

Hard · stretch and reason

7 PROBLEMS

1. Solve $x^2 + \frac{1}{x^2} = 2.5$

ANSWER $x = \pm 2, \pm 0.5$

2. Solve $\frac{x}{x+1} + \frac{x+1}{x} = 2.5$

ANSWER $x = 1, -2$

3. Solve $\frac{1}{x^2-1} + \frac{1}{x-1} = 1$

ANSWER $x = 2$ or $x = -1$ rejected

4. Solve $\frac{2x}{x^2-4} + \frac{1}{x-2} = \frac{3}{x+2}$

ANSWER $x = -8$

5. Find k so $\frac{x}{x-2} = k$ has no solution

ANSWER $k = 1$

6. Solve $\frac{x^2-4}{x^2-2x} = 1$

ANSWER no solution

7. Solve $(x + \frac{1}{x})^2 - 4(x + \frac{1}{x}) + 3 = 0$

ANSWER $x = 1$ (from $t = 3: \frac{3 \pm \sqrt{5}}{2}$)

07 Homework

Homework — 5 tasks

Every solution must open with the domain and close with the check.

1. Solve $\frac{3}{x} + \frac{2}{x-1} = 3$.
2. Solve $\frac{x}{x-4} = \frac{2}{x+1}$.
3. Solve $\frac{x^2}{x+3} = \frac{9}{x+3}$ and say which root is rejected and why.
4. Solve $\frac{1}{x} + \frac{1}{x+2} = \frac{3}{4}$.
5. Solve $x^2 + \frac{4}{x^2} = 5$ using a substitution.